

AI Resume Copilot - Complete SRS & Project Blueprint

1. Executive Summary

AI Resume Copilot allows users to create one master career profile and automatically generate multiple ATS-optimized resumes tailored to specific job descriptions. The platform acts as an AI Career Copilot rather than a traditional resume builder.

2. Target Users

• Students applying for internships • Software Engineers • Data Engineers • AI/ML Engineers • Experienced professionals changing jobs • Career coaches and placement cells

3. Functional Requirements

FR-1 User Authentication FR-2 Profile Management FR-3 Resume Template Selection FR-4 JD Upload / Paste FR-5 AI Resume Tailoring FR-6 ATS Score Generation FR-7 Skill Gap Analysis FR-8 Resume Versioning FR-9 Cover Letter Generation FR-10 Interview Question Generation FR-11 Resume Review Agent FR-12 PDF Export

4. Non Functional Requirements

• Response time < 15 sec • Secure authentication • Scalable architecture • Mobile responsive UI • Version history • Audit logging

5. Detailed Database Schema

Users - id - name - email - created_at Education - id - user_id - institute - degree - cgpa - start_date - end_date Skills - id - user_id - skill_name - category Projects - id - user_id - title - description - tech_stack - github_link Experience - id - user_id - company - role - description Achievements - id - user_id - content ResumeVersions - id - user_id - jd_id - resume_json - pdf_url ATSReports - id - resume_id - score - missing_skills

6. ER Relationship Overview

User → Education (1:M) User → Skills (1:M) User → Projects (1:M) User → Experience (1:M) User → Achievements (1:M) User → ResumeVersions (1:M) ResumeVersion → ATSReport (1:1)

7. Frontend Architecture

Framework: Next.js 15 Language: TypeScript UI: Tailwind + ShadCN Pages: /login /dashboard /profile /resumes /jd-analysis /reviewer /settings Components: Navbar ProfileForm ResumePreview ATSScoreCard JDAnalyzer PDFExporter

8. Backend Architecture

Framework: FastAPI Modules: api/ services/ database/ prompts/ models/ auth/ Services: Resume Generator ATS Engine JD Analyzer Reviewer Agent Cover Letter Agent Interview Agent

9. API Design

POST /auth/login POST /profile/create GET /profile POST /resume/generate POST /resume/review POST /ats/score POST /cover-letter/generate POST /interview/questions GET /resume/history

10. AI System Design

Input: User Profile + Job Description Pipeline: 1. Extract JD requirements 2. Skill matching 3. Project ranking 4. Experience ranking 5. Achievement rewriting 6. Resume JSON generation 7. ATS analysis 8. PDF rendering

11. RAG Architecture

Embedding Model: text-embedding-3-small or Gemini Embeddings Vector Database: Qdrant Flow: Profile Data → Embeddings → Qdrant → Relevant Context Retrieval → LLM → Resume Generation

12. ATS Scoring Algorithm

Score Components: 40% Skills Match 25% Experience Match 15% Keyword Match 10% Project Relevance 10% Formatting Output: ATS Score Missing Skills Improvement Suggestions

13. Prompt Engineering Strategy

Rules: • Never invent experience • Never add fake skills • Reorder content only • Rewrite for clarity • Optimize ATS keywords • Output structured JSON

14. Resume JSON Structure

```
{ summary:{}, skills:[], projects:[], experience:[], education:[], achievements:[] }
```

15. Resume Templates

Template 1: Software Engineer Template 2: AI Engineer Template 3: Data Engineer Template 4: Research Resume Template 5: Modern ATS Resume

16. AI Reviewer Agent

Outputs: Resume Score Strengths Weaknesses ATS Risk Factors Improvement Recommendations

17. Interview Agent

Input: Resume + JD Output: Likely Technical Questions Behavioral Questions System Design Questions Role Specific Questions

18. Cover Letter Agent

Generate: Cover Letter LinkedIn DM Referral Request Application Email

19. Deployment Architecture

Frontend: Vercel Backend: Railway / Render / AWS ECS Database: Supabase PostgreSQL Vector DB: Qdrant Cloud Monitoring: Sentry PostHog

20. Security

JWT Authentication HTTPS Rate Limiting Prompt Injection Protection Input Validation Audit Logs

21. Cost Estimation

MVP: Frontend: Free Backend: \$10-20/month Database: Free Tier OpenAI API: \$10-30/month
Total: ~\$20-50/month

22. Development Timeline

Week 1: Authentication + Database Week 2: Profile Management Week 3: AI Resume Generation
Week 4: ATS Engine Week 5: Reviewer + Interview Agent Week 6: Deployment + Testing

23. Resume Project Description

Built an AI-powered ATS Resume Optimization Platform that generates job-specific resumes from a single master profile using LLMs, RAG, semantic search, ATS scoring, and automated PDF generation. Implemented FastAPI, Next.js, PostgreSQL, Qdrant, and OpenAI APIs to deliver personalized career assistance and interview preparation.

24. Final Positioning

Product Name: AI Resume Copilot Tagline: One Profile. Unlimited Job-Specific Resumes. Recruiter Value: 9.5/10 Portfolio Project Skills Demonstrated: Full Stack Development AI Engineering RAG Vector Databases Prompt Engineering System Design Product Thinking